

Jiayi Guo

Biography

Jiayi Guo is currently a Ph.D. student in Mathematics with a concentration in Actuarial and Risk Management Sciences at the University of Illinois Urbana-Champaign (UIUC). Prior to beginning her doctoral studies, she earned a Master's degree in Predictive Analytics and Risk Management from UIUC. Her research interests lie in actuarial science, climate risk, catastrophe risk modeling, and data-driven risk assessment. She is particularly interested in quantifying the impacts of extreme weather events on insurance losses and developing statistical methods for risk measurement and prediction. Jiayi currently serves as a graduate supervisor for undergraduate research projects at the Illinois Risk Lab. Her recent work focuses on climate-related insurance risk, catastrophe loss modeling, and data preparation methodologies for actuarial applications. She has presented her research at multiple Actuarial Research Conferences and actively contributes to projects that advance the use of climate and loss data in insurance and risk management.

Education

- 2024–Present **Ph.D. Student in Mathematics, Concentration in Actuarial and Risk Management Sciences**, *University of Illinois Urbana-Champaign*, Champaign, USA, Advisor: Professor Zhiyu Quan, PhD
- 2022–2023 **M.S., Predictive Analytics and Risk Management**, *University of Illinois Urbana-Champaign*, Champaign, USA
- 2018–2022 **B.S., Statistics**, *Dalian Maritime University*, Dalian, China

Research Interest

Actuarial Science, Catastrophe Risk Modeling, Insurance Loss Analytics

Professional Exams

SOA (Society of Actuaries): Exam P, FAM(UEC), FM (UEC)

Professional Experience

- 2024–Present **Graduate Supervisor**, *University of Illinois Urbana-Champaign*, Illinois Risk Lab, USA
- 2024–Present **Graduate Research/Teaching Assistant**, *University of Illinois Urbana-Champaign*, Department of Actuarial and Risk Management Sciences, USA

Publications

Refereed Journal Articles

- 2026 **Jiayi Guo**, Linfeng Zhang, Zhiyu Quan. (2026). Entity-specific cyber risk assessment using InsurTech-empowered risk factors *Variance, Risk Management*, 19. <https://doi.org/10.66573/001c.155715>
- 2022 Yunhui Pang, Lidong Wang, Yifei Liu, and **Jiayi Guo**. (2022). Fuzzy rule-based models via space partition and information granulation. *Neural Computing and Applications*, 34(19), 16199–16211. <https://doi.org/10.1007/s00521-022-06974-3>

Working Manuscripts

- 2026 **Jiayi Guo**, Panyi Dong, Zhiyu Quan.(2026). Starting off on the wrong foot: pitfalls in data preparation. Submitted for review. <https://arxiv.org/abs/2603.18190>

Awards & Honors

- 2026 **PARM Summer Research Fellows**, *University of Illinois Urbana-Champaign*
2025 **PARM Summer Research Fellows**, *University of Illinois Urbana-Champaign*

Presentations

- 2026 **Starting Off on the Wrong Foot: Pitfalls in Data Preparation.**, *29th International Congress on Insurance: Mathematics and Economics (IME 2026)*, Sungkyunkwan University, Seoul, South Korea, Jun. 29-Jul. 3, 2026
- 2025 **Entity-Specific Cyber Risk Assessment using InsurTech Empowered Risk Factors.**, *60th Actuarial Research Conference (ARC)*, York University, Toronto, Canada, Jul. 29-Aug. 1, 2025
- 2023 **Estimating the Climate-Change Impact on a Fixed-Income Investment Portfolio due to Physical and Transitional Risks**, *58th Actuarial Research Conference (ARC)*, Drake University, Des Moines, USA, Jul. 30-Aug. 2, 2023
- 2021 **Active Learning Method Based on Axiomatic Fuzzy Sets and Cost-Sensitive Classification.**, *International Conference on Neural Computing for Advanced Applications (NCAA)*, Guangzhou, China, Aug. 27-30, 2021

Research Experience

- 2025-Present Automated Machine Learning in Insurance
- Spring 2025 Entity-Specific Cyber Risk Assessment using InsurTech Empowered Risk Factors
- Spring 2023 Estimating the Climate-Change Impact on a Fixed-Income Investment Portfolio due to Physical and Transitional Risks
- Graduate Supervisor, Illinois Risk Lab
- 2024-Present Climate Data Innovation and Consolidation Projects with SOA Collaboration

Teaching Experience

Teaching Assistant, University of Illinois Urbana-Champaign

- Spring 2026 **ASRM 555 Advanced Predictive Analytics**, *Course Development, Grading*
ASRM 469 Casualty Actuarial Mathematics, *Course Development, Grading, Office Hour*
- Fall 2025 **ASRM 455 Predictive Analytics**, *Course Development, Grading, Office Hour*
- Spring 2025 **ASRM 555 Advanced Predictive Analytics**, *Course Development, Grading*
ASRM 469 Casualty Actuarial Mathematics, *Course Development, Grading, Office Hour*
- Fall 2024 **ASRM 455 Predictive Analytics**, *Course Development, Grading, Office Hour*